Third Terminal Examination - 2082

Class : XI | **Subject: Chemistry (3011)** | **Full Marks: 75**

Time: 3 hrs. | **SET - A** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. The number of molecules in 36 mg of water is
   a. 12×10^{23} b. 6.023×10^{23}
   c. 12.04×10^{20} d. 12×10^{21}
2. The hybridization shown by Nitrogen in ammonium ion (NH_4^+) is
   a. sp b. sp^2 c. sp^3 d. sp^3d^2
3. Which of the following has highest percentage of hydrogen?
   a. CH_4 b. C_2H_4 c. C_6H_6 d. C_2H_6
4. Which of the following properties of water can be used to explain the spherical shape of rain droplets?
   a. surface tension b. viscosity
   c. boiling point d. density
5. At constant temperature, when the product of pressure and volume is plotted against pressure for a given amount of gas, the line obtained is:
   a. parallel to x-axis b. parallel to y-axis
   c. a rectangular hyperbola d. linear with a positive slope
6. SO_3 gas is formed as an intermediate during the manufacture of Sulphuric acid by contact process. The formation of Sulphur trioxide from sulfur dioxide and oxygen is reversible.
   $2SO_2 + O_2 \rightleftharpoons 2SO_3 \; \Delta H = -196 \text{ kJ mol}^{-1}$
   Which conditions of pressure and temperature favor the reverse reaction?
   a. High pressure and high temperature
   b. High pressure and low temperature
   c. Low pressure and high temperature
   d. Low pressure and low temperature

7. Which is the purest form of carbon?
   a. Diamond
   b. Graphite
   c. Anthracite
   d. Charcoal
8. Liquid helium is due to the
   a. Ionic bonds
   b. Covalent bonds
   c. Van der Waals forces
   d. Hydrogen bonds
9. Aqua-regia is
   a. 1 Conc. HNO_3 + 2 Conc. HCl
   b. 1 Conc. HNO_3 + 3 Conc. HCl
   c. 1 Conc. HNO_3 + 1 conc. HCl
   d. 3 Conc. HNO_3 + 1 Conc. HCl
10. Which of the following is an optically active compound?
    a. 1-chloropropane
    b. 2-chloropropane
    c. 1-chlorobutane
    d. 2-chlorobutane
11. The pyrolysis of n-pentane produces
    a. ethane and ethene
    b. propene and ethane
    c. propene and propyne
    d. n-butane and methane

Group – B

Give short answer to the following questions. **[8 × 5 = 40]**

12. Neil Bohr proposed a new model of the atom in 1913 to overcome the shortcomings of Rutherford's model and to explain the line spectrum of hydrogen. He incorporated Planck's quantum theory and suggested that electrons revolve around the nucleus in specific permitted circular paths without radiating energy.
    a. Write all the postulates of Bohr’s atomic model. [3]
    b. Draw and label α-particle scattering experiment. [2]
13. Define redox reaction. Balance the redox reaction with the help of oxidation number method or ion electron method. Also, identify oxidant and reductant. [1+3+1]

$$NaOH + Cl_2 \longrightarrow NaCl + NaClO_3 + H_2O$$

OR

Electrolysis is a process of chemical decomposition of an electrolyte in solution by the passage of electric current.

a. State Faraday's second law of electrolysis. [1]

b. Complete the following table [2]

Electrolyte	Electrode	Anode deposit	Cathode deposit
aqueous NaCl	Inert electrode	Cl_2	------------------
Aq. $AgNO_3$	Inert electrode	------------------	Ag

c. 1.5 gram of a trivalent metal (M) was deposited at cathode by passing a current of 2.5A through its salt solution for 30 min. What is the atomic mass of metal (M)? [2]

14. Valence shell electron pair repulsion (VSEPR) theory predicts molecular geometry based on electron pair repulsion. Whereas, valence bond theory (VBT) explains bond formation through orbital overlapping.

a. Differentiate between sigma and pi bond. [1]

b. Complete the given table indicating shape, bond angle and hybridization of molecule. [2]

Molecule	Shape	Bond angle	Hybridization
NH_3	Pyramidal	107°	sp^3
H_2O			
BF_3			

c. Define hydrogen bonding. Write its type with one example of each. [2]

15. Inorganic Chemistry deals with the study of metals, non-metals, acids, bases, salts, and their reactions, properties, and preparation

a. What are the differences between atomic hydrogen and nascent hydrogen? [2]

b. Describe with one chemical reaction to show that nascent hydrogen is more reactive than molecular hydrogen. [1]

c. Why is Al_2O_3 an amphoteric oxide? Explain with chemical reactions. [1]

d. What is passivity of iron? Why is it called so? [1]

16. Ozone is a highly reactive gas composed of three oxygen atoms. It is both a natural and a man-made product that occurs in the stratosphere and lower atmosphere.

a. How is ozone formed in stratosphere? [1]

b. How does chlorofluorocarbon deplete the ozone layer? Explain with chemical reactions. [2]

c. What is tailing of mercury? [1]

d. Mention the major uses of ozone. [1]

17. Carbon and phosphorus are exist in free state as well as in combined state. In free state they show phenomena of allotropy.

a. Define allotropy. [1]

b. Diamond is a bad conductor, but graphite is a good conductor of electricity. Explain. [2]

c. How does carbon monoxide reacts with Nickel? [1]

d. What happens when phosphine gas is passed into aq. solution of $CuSO_4$? [1]

18. Organic compounds can be grouped into families based on their similar structures and properties. These families show a regular pattern in their composition and chemical behavior.

a. Define homologous series. [1]

b. Write the major characteristics of homologous series. [2]

c. Write the formula and IUPAC name of first member of alcohol and third member of alkene. [2]

19. According to the IUPAC system, the name of organic compound is made up of different parts.

a. Write the structure of the following IUPAC name. [2]

i. 2-chloro-4-methylpent-2-ene

ii. 2-methylpropan-1-ol

b. Write the IUPAC name of following compounds [3]

i.
$$\begin{array}{ccccccc} & & CH_3 & & & & \\ & & | & & & & \\ CH_3 & - & CH & - & CH & - & CH_3 \\ & & & & | & & \\ & & & & CH_3 & & \end{array}$$

ii. $CH_3 - CH_2 - CH_2 - \overset{\overset{O}{||}}{C} - OH$

iii. $CH_3 - CH_2 - \overset{\overset{O}{||}}{C} - O - CH_3$

Group - C

Give long answer to the following questions. **[3 × 8 = 24]**

20. The kinetic theory of gases explains the properties of gases, like volume, pressure, and temperature, along with transport properties such as viscosity and thermal conductivity. It also accounts for phenomena, like Brownian motion.

    a. Write the main postulates of Kinetic theory of gas. [3]

    b. Derive the ideal gas equation PV = nRT where the symbols have their usual meaning. [2]

    c. A vessel of volume 100ml contains 10% O_2 and 90% unknown gas. The gases diffuse in 86 sec. through a small hole of the vessel. If pure oxygen under same condition diffuses in 75 sec., Find the molecular mass of the unknown gas? [2]

    d. Differentiate between efflorescence and hygroscopic with example. [1]

21. Both Sulphur dioxide (SO_2) and Chlorine (Cl_2) act as good bleaching agents.

    a. Differentiate between bleaching action of Cl_2 and SO_2. [2]

    b. What happens when chlorine is heated with hot and conc. NaOH? [1]

    c. Write the reaction of iodine and ammonia. [1]

    d. How would you test the presence of Cl^- ion in its aqueous solution? [1]

e. What happens when SO_2 is passed through acidified $FeCl_3$ solution? [1]

f. What happens when H_2S is passed through acidified $K_2Cr_2O_7$ solution? [1]

g. Mention an important application of each of the following. [1]

i. Hypo ii. Heavy water

OR

Sulfuric acid is one of the largest volumes of industrial chemical produced in the world. Over the last decades the contact process has been used to produce sulfuric acid, replacing the traditional (Lead Chamber) process.

a. Write the four steps of chemical equations for the manufacturing of sulphuric acid by contact process starting from iron pyrite. [4]

b. Why is SO_3 not directly dissolved in water to prepare sulphuric acid? [1]

c. Give any two chemical equations in which sulphuric acid acts strong acid and oxidizing agent. [2]

d. What is charring action? Describe with examples. [1]

22. Organic chemistry deals with the structure, properties, and analysis of carbon compounds. Different structural arrangements and laboratory techniques help in understanding and identifying various organic substances.

a. Define isomerism. [1]

b. Write the positional isomers of compound having molecular formula C_3H_8O with their IUPAC name. [2]

c. What are Geometrical isomers? Give suitable example of it. [2]

d. Why is it necessary to prepare sodium extract during the detection of foreign element? [1]

e. Write the reaction involved in the detection of nitrogen. [2]

The End